Vishwas K Singh
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Python Training Day 1

Agenda

Fundamental Terms and Concepts
Installing and setting up python
Literals, Variables, and Numeral Systems
Input and Output in Python (Console I/O)
Operators and Expressions

Timings

9.30 to 11.00AM – p1
11 – 11.15 – break
11.15 to 1.30 – p2
1.30 to 2.30 – lunch break
2.30 to 3.45 – p3
3.45 to 4.00 – break
4.00 to 5.30 – p4

Have you ever wanted a solution

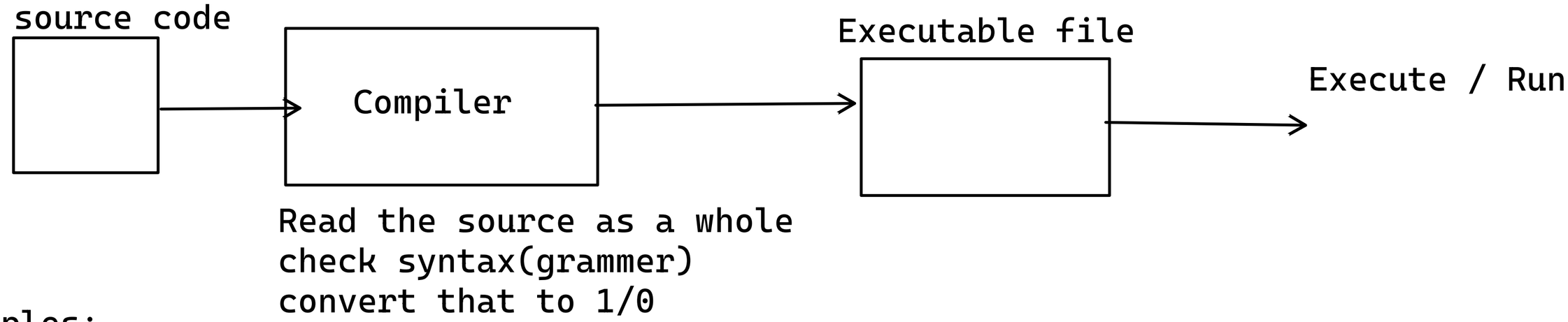
where you get data from various departments
you have to compile those, generate insights, manualy create a report out of it

to automate this?

Programming language
your High level English kind of language to machine instructions

There are 2 kinds of languages
compiled language
interpreted language

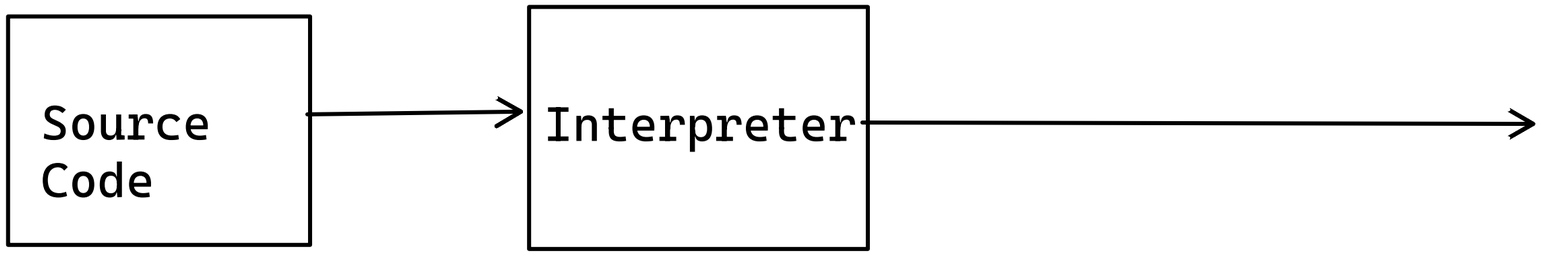
Compiled Language



Read the source as a whole
check syntax(grammer)
convert that to 1/0

Examples:
c, C++, Csharp

Interpreted Language



Examples:
Python
Javascript

Script

Read each command (statement)
Convert that to 1/0
execute the command/ run the command

Python Programming Language — Guido Van Rossum

1. General purpose Language
2. Open Source language
3. Architecture Independent Language
4. Dynamically Typed Language (later)

1. I can perform - Web programming, system applications, Numerical Computation

2. Open Source Language:
- Source Code is available, for free to use, modify & redistribute
- Example: Linux (ubuntu, red hat, fedora), Android phone

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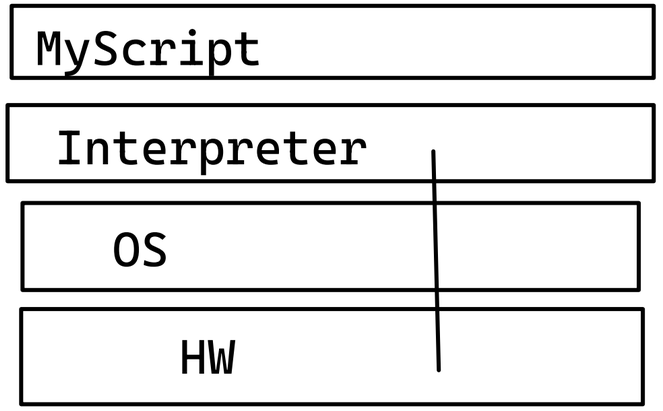
Architecture Independence

MyScript can be run on any architecture without modifications



Windows 64bit	Windows 64	any other os/ any spec
Intel	AMD	other

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Java / Dotnet vs Python

- Its almost similar
- Python actually takes features from c, c++, Java
- they are Object oriented Language, but python is Functional programming Language

Interpreter of Python
- Needed to run a Python Script

Versions
- C Cython C Code + Python can be used
- Java Jython Java Code + Python
- R RPython R Code + Python
- Python Pypi

Versions of Python
2.x.x
3.x.x (latest recommended)

2.x ~~3.x~~ python.org

3.9 — 3.12 — 3.15

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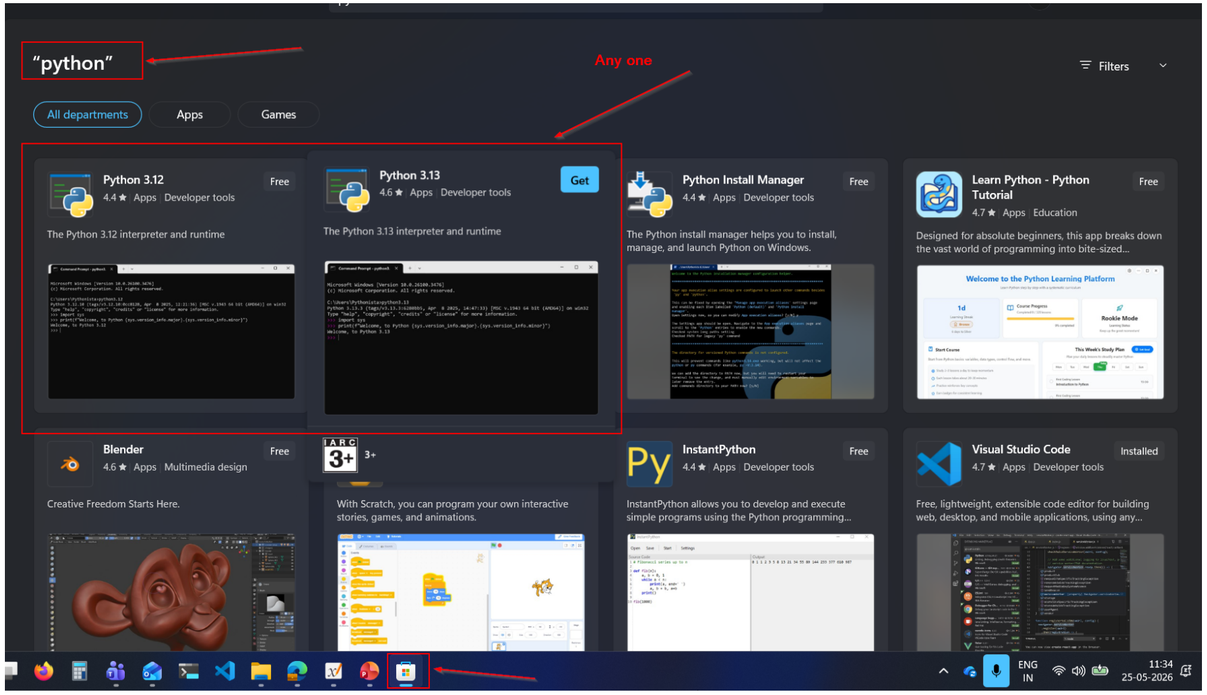
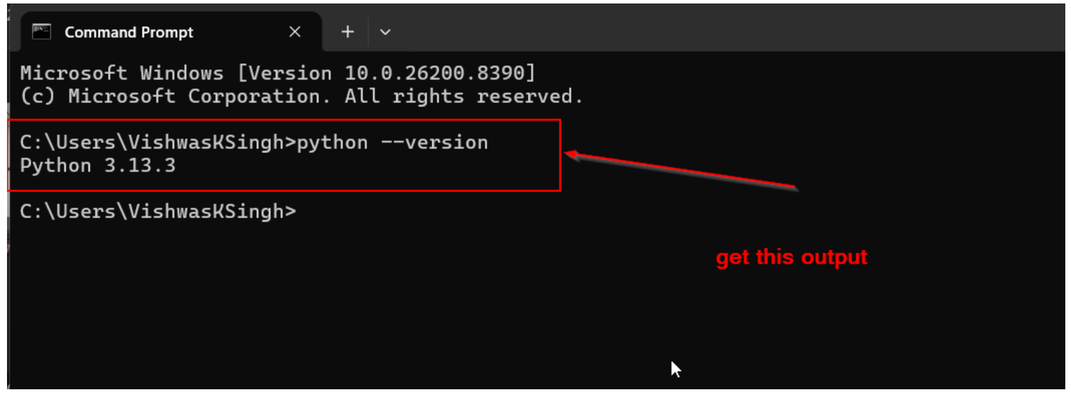
Lab: Installation of Python





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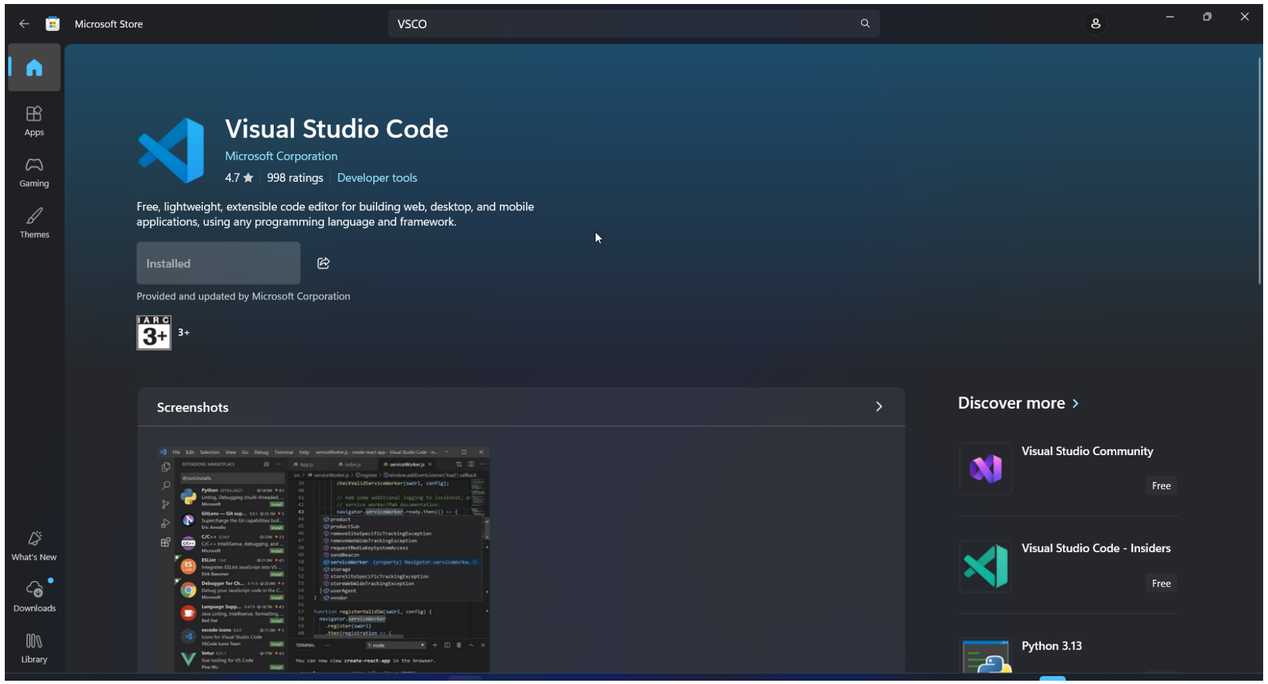
Lab: Text Editor: VS Code



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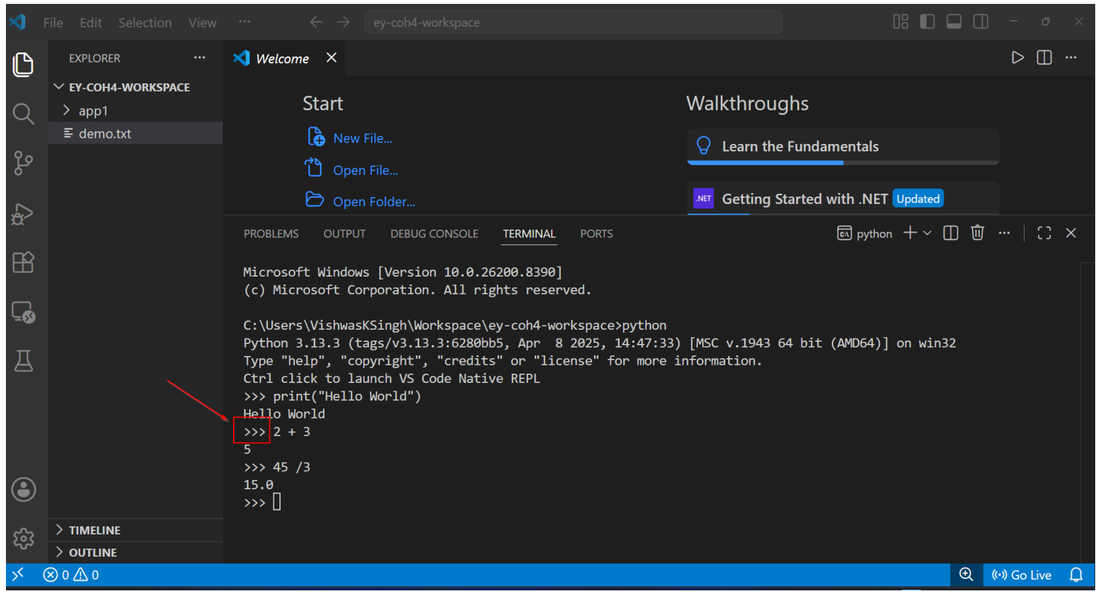
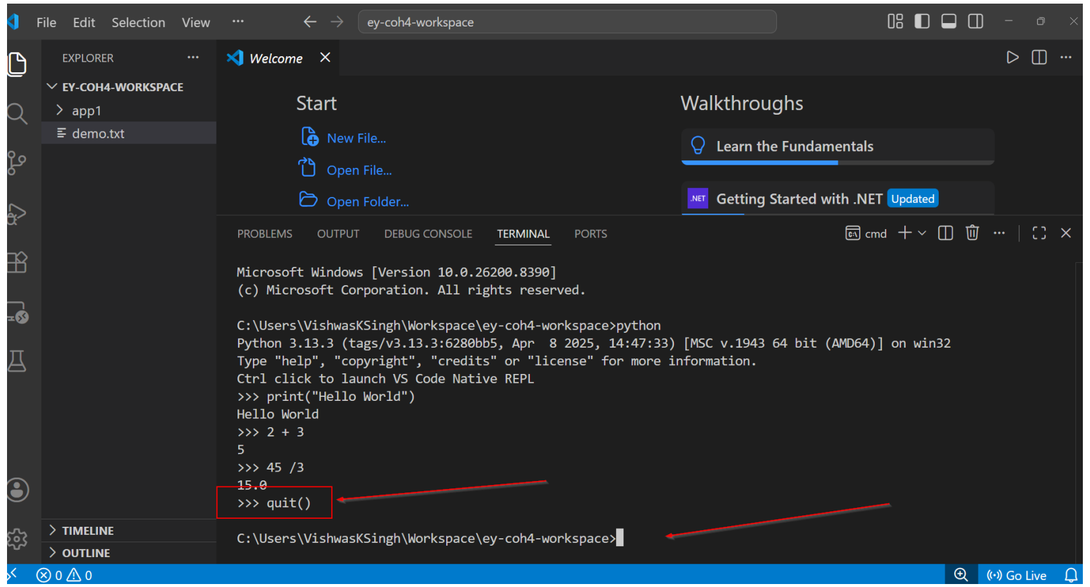
Interacting with Python

1. Interactive Mode / Prompt mode





We use this when we are initially creating, or we need to understand working of a small code snippet

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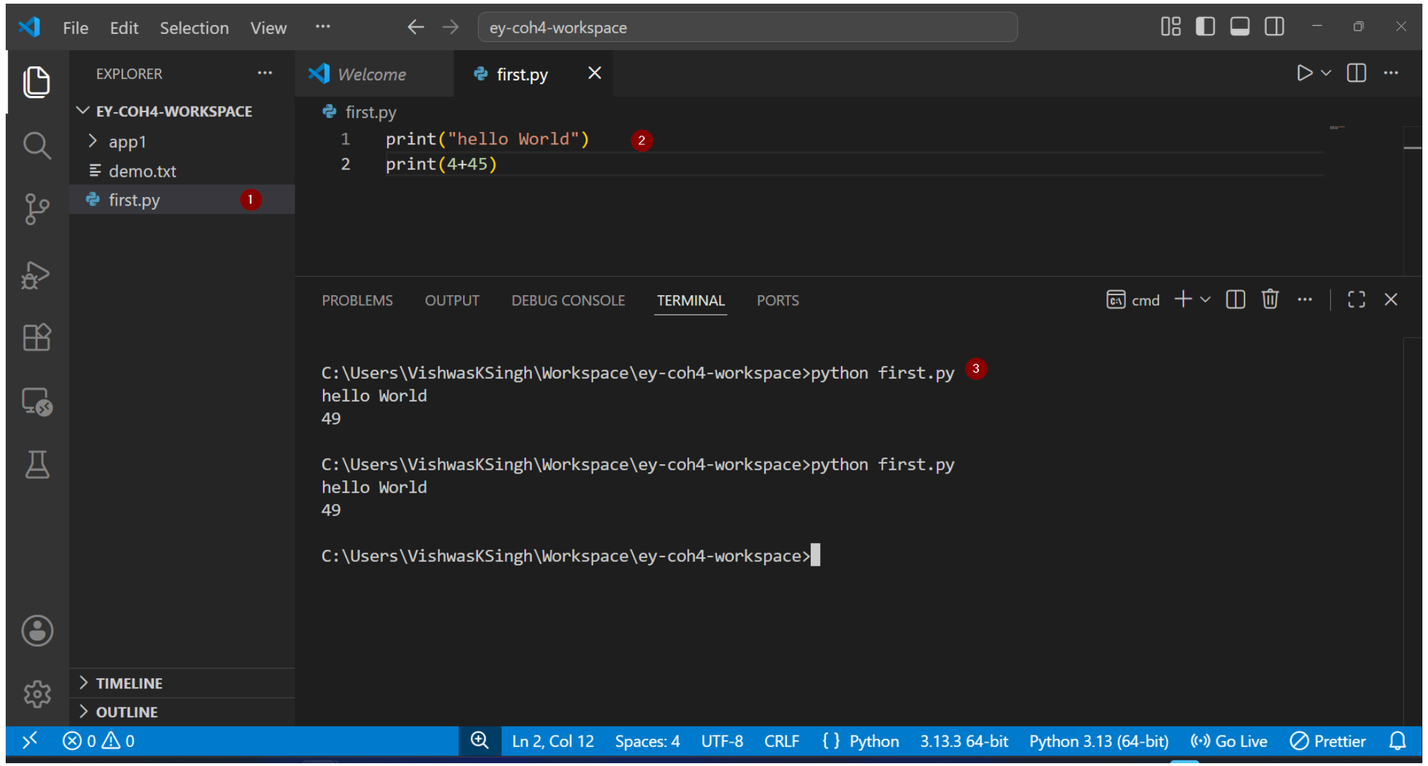
2. Script Mode of Execution



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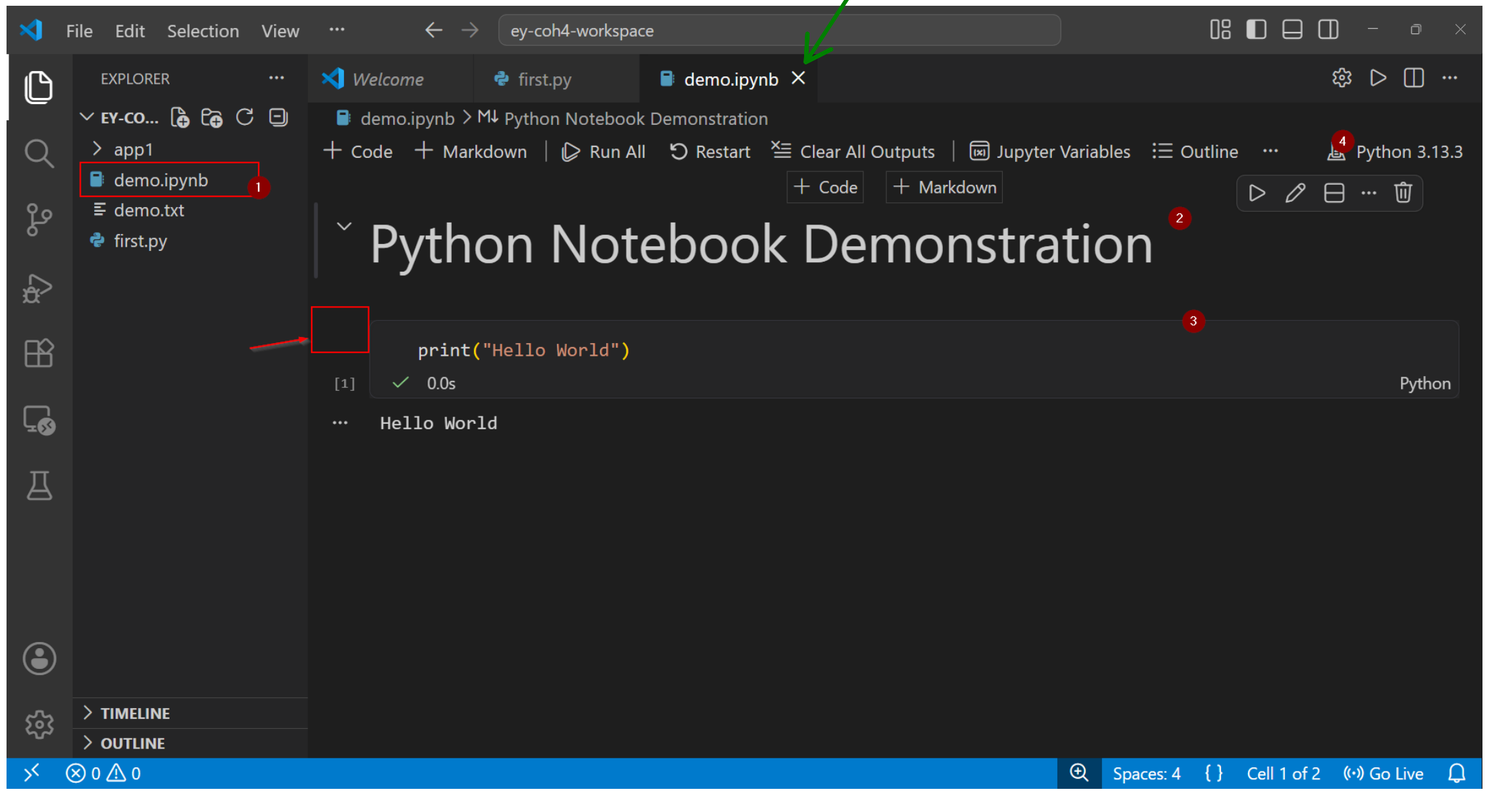
3. Notebook Mode (.ipynb)

Notes + Interactive + saved
Data Analyst/ Scientist

Jupyter Notebooks
Google Colab



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Python Language Basics

Learning how to program

When ever you are given a problem
Break the problem to sub problem
Find Solution to each sub problem
Integrate it

→

Analyse the sub problem
Input?
Output?
What is relationship b/n Input & Output?
(Also called Logic)

Example:
- get data from various departments
- you have to compile those
- generate insights
- create a report out of it

→

What format is your input → Eg. Excel File
- Read the Excel File(Input)
- Clean It (Logic process)
- Format for next processing (output)

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To Learn a Programming Language

1. How to store data temporarily
2. How to do Conditions
3. How to repeat (Loops and Functions)
4. What the base paradigm

   Python is a object based Functional Programming Language

What are alternatives to Functional Programming Paradigm?
O - Object oriented Programming
P - Procedural Programming Approach

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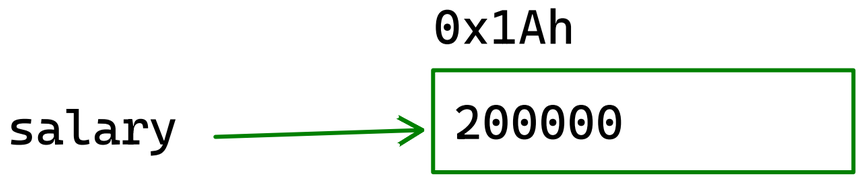
1. How to store data temporarily

Variables
Label to memory location

```
salary          =          2000000
```

- salary: variable identifier
- =: Assignment operator
- 2000000: data



- It can contain alphanumeric characters, underscore(_)
- It must always start with a letter or _
- no other special characters

- must not be a keyword
- meaningful (good practice)
- It is case-sensitive

```
C:\Users\VishwasKSingh>python
Python 3.13.3 (tags/v3.13.3:6280b
Type "help", "copyright", "credit
>>> salary = 20000
>>> salary
20000
>>> print(salary)
20000
>>> type(salary)
<class 'int'>
>>> name = "vishwas"
>>> type(name)
<class 'str'>
>>> bonus = 2135.6789
>>> type(bonus)
<class 'float'>
>>>
```

Keywords

```
False
None
True
and
as
assert
async
await
break
class
continue
def
del
elif
else
except
finally
for
from
global
if
import
in
is
lambda
nonlocal
not
or
pass
raise
return
try
while
with
yield
```

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Data Type
- Int (1234)
- Float (12.34)
- String ("12.34")
- Bool (True)
- Complex (2+3j)

```
>>> k = 1234
>>> k
1234
>>> type(k)
<class 'int'>
>>> k1 = 12_00_234
>>> type(k1)
<class 'int'>
>>> k2 = 0b1010
>>> k2
10
>>> type(k2)
<class 'int'>
>>> k3 = 0x0A
>>> k3
10
>>> type(k3)
<class 'int'>
>>> k4 = 0o27
>>> k4
23
>>> type(k4)
<class 'int'>
>>> k5 = complex(2,3)
>>> k5
(2+3j)
>>> type(k5)
<class 'complex'>
>>>
```

```
>>> f1 = 12.3456
>>> type(f1)
<class 'float'>
>>> f2 = 123456e-4
>>> f2
12.3456
>>> type(f2)
<class 'float'>
>>>
```

```
>>> f3 = True
>>> f3
True
>>> type(f3)
<class 'bool'>
>>> f4 = False
>>> type(f4)
<class 'bool'>
>>> f4
False
>>>
```

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```
>>> s1 = "vishwas"
>>> s1
'vishwas'
>>> type(s1)
<class 'str'>
>>> s2 = 'vishwas'
>>> s2
'vishwas'
>>> type(s2)
<class 'str'>
>>> s3 = '''vishwas'''
>>> s3
'vishwas'
>>> type(s3)
<class 'str'>
>>> s4 = """vishwas"""
>>> s4
'vishwas'
>>> type(s4)
<class 'str'>
>>> s5 = "This is Vishwas's book"
>>> s6 = '''
... This is an example of
... multiline value. It is considered
... as a single string itself
... '''
>>> s6
'\nThis is an example of\nmultiline value. It is considered\nas a single string itself\n'
>>>
```

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```
>>> s7 = "\nI am Vishwas K Singh\n I am Learning python"
>>> s7
'\nI am Vishwas K Singh\n I am Learning python'
>>> print(s7)

I am Vishwas K Singh
 I am Learning python
>>> s8 = "C:\Users\VishwasKSingh\Workspace\ey-coh4-workspace\first.py"
  File "<python-input-65>", line 1
    s8 = "C:\Users\VishwasKSingh\Workspace\ey-coh4-workspace\first.py"
         ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
SyntaxError: (unicode error) 'unicodeescape' codec can't decode bytes in position 2-3: truncated \UXXXXXXXX escape
>>> s8 = "C:\\Users\\VishwasKSingh\\Workspace\\ey-coh4-workspace\\first.py"
>>> s8
'C:\\Users\\VishwasKSingh\\Workspace\\ey-coh4-workspace\\first.py'
>>> print(s8)
C:\Users\VishwasKSingh\Workspace\ey-coh4-workspace\first.py
>>> s8
'C:\\Users\\VishwasKSingh\\Workspace\\ey-coh4-workspace\\first.py'
>>> s9 = r"C:\Users\VishwasKSingh\Workspace\ey-coh4-workspace\first.py"
>>> s9
'C:\\Users\\VishwasKSingh\\Workspace\\ey-coh4-workspace\\first.py'
>>> name = "vishwas"
>>> greet = f"Hey there!, Welcome to session {name}"
>>> greet
'Hey there!, Welcome to session vishwas'
>>> version = 3
>>> greet = f"Hey there!, Welcome to session {name}, I am learning python {version}"
>>> greet
'Hey there!, Welcome to session vishwas, I am learning python 3'
>>> salary = 123.4567890
>>> greet = f"Hey there!, Welcome to session {name}, I am learning python {version}, I am earning a salary of {salary:.2f}"
>>> greet
'Hey there!, Welcome to session vishwas, I am learning python 3, I am earning a salary of 123.46'
>>>
```

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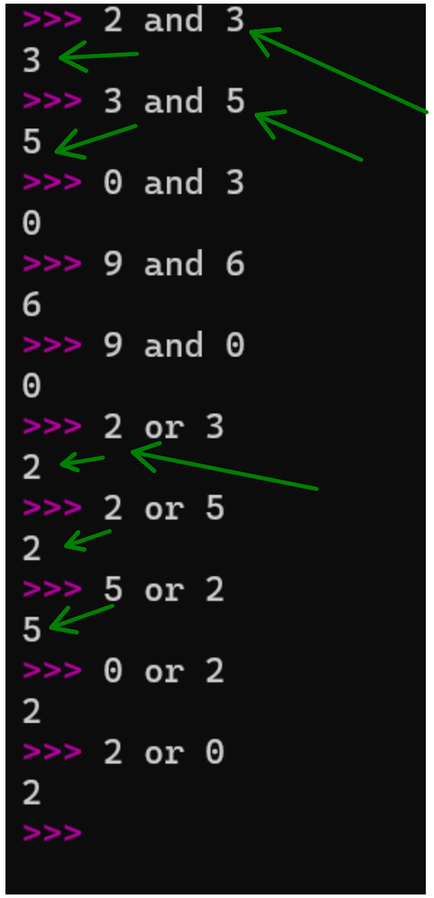
Operators

- Assignment (=,+=,-=,*=....)
- Arithmetic (+,-,*,/,%, //(integer division), ** (power))
- Logical (and,or,not)
- Relational (<,>,≤,≥,≠)
- Identity (as)
- Membership (in,not in)
- Bitwise (&,|,~)

Pointers:
- A 0(zero) is always False
- Any non-zero number is always True

For faster evaluation python uses "Short Ciruit Evaluation"
- In a logical expression expr1 and expr2
  expr2 is not evaluated unless expr1(True)
- In a logical expression expr1 or expr2
  expr2 is not evaluated at all when expr1 is already True



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"Guard Pattern"

```
>>> x = 10
>>> y = 2
>>> x / y
5.0
>>> y = 0
>>> x / y
Traceback (most recent call last):
  File "<python-input-94>", line 1, in <module>
    x / y
    ~~^~~
ZeroDivisionError: division by zero
>>> y != 0 and x / y
False
>>> y = 2
>>> y != 0 and x / y
5.0
>>>
```

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Lab: GST Calculation v1.0

The GST on Electronic Applicances was reduced from 28% to 18%. If an AC costed Rs.30000(excluding GST) what are the values of GST at both the rates & what is the difference in cost (how much you would save now)?

Assume no other taxes are applied

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Console I/O in python
- printing to a conole - print(....)
- taking values from a console - input(....)

Input Prompt
i.e message for the user

Pointers:
1. Whatever you take from an input function is always a string
2. You as a developer have to convert to right data type

```
>>> age = input("Enter your age: ")
Enter your age: 56
>>> age
'56'
>>> type(age)
<class 'str'>
>>> age = int(age)
>>> age
56
>>> type(age)
<class 'int'>
>>>
```

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Lab: GST Calculation v2.0

gst_calcv2.py > ...

```python
# Input Section
ac_cost = float(input("Enter the AC Price: "))
old_gst = float(input("Enter old GST Rate: ")) / 100 # Calculating actual value for 28%
new_gst = float(input("Enter new GST Rate: ")) / 100

# Logic or calculation
old_total_cost = ac_cost + (ac_cost * old_gst)
new_total_cost = ac_cost + (ac_cost * new_gst)

total_savings = old_total_cost - new_total_cost

# Output
print(f"The Cost Saving of a AC at Rs.{ac_cost} with reduction in GST from {old_gst * 100:.2f}% to {new_gst * 100}%\
 is Rs.{total_savings} ")

```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
C:\Users\VishwasKSingh\Workspace\ey-coh4-workspace>python gst_calcv2.py
Enter the AC Price: 25000
Enter old GST Rate: 28
Enter new GST Rate: 20
The Cost Saving of a AC at Rs.25000.0 with reduction in GST from 28.00% to 20.0% is Rs.2000.0
```

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Summary Day 1

We learnt about

1. What is programming How to approach it
2. Different kinds of Programming Language
3. Features of Python Programming Language, different versions, different interpreters
4. Variables, Data Types & Operators, Console I/O in python

What to look out for tomorrow?

- Conditionals
- Loops
- Strings, Lists, Dicts, Tuples, Sets

****